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CONSCIOUSNESS AND SLEEP

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Summary

Sleep is a universal, essential biological process. It is also an invaluable window on consciousness. It tells us that consciousness can be lost, but also that it can be regained, in all its richness, when we are disconnected from the environment and unable to reflect By considering neurophysiological differences between dreaming and dreamless sleep, we can learn about the substrate of consciousness and understand why it vanishes. We also learn that the ongoing state of the substrate of consciousness determines the way each experience feels regardless of how it is triggered—endogenously or exogenously. Dreaming consciousness is also a window on sleep and its functions. Dreams tell us that the sleeping brain is remarkably lively, recombining intrinsic activation patterns from a vast repertoire, freed from the requirements of ongoing behavior and cognitive control.

In brief:

The paper by Tononi et al. discusses what sleep reveals about the substrate of consciousness, how experience can vanish during sleep, and how dreams can be supported by a brain largely disconnected from the environment, unreflective, and forgetful.

Introduction: Sleep as a window on consciousness

Perhaps the most important experiment ever done is one we perform every night: fall asleep and reawaken. This simplest of experiments reveals two fundamental truths. First, consciousness can be lost, and when it is lost there is nothing at all—from one's intrinsic perspective, the universe has vanished. Second, consciousness—one's private universe—can be regained in dreams, even though we are disconnected from the universe outside.

The recognition that consciousness can be lost during sleep underlies traditional representations of sleep as the "brother of death." On the other hand, the return of consciousness underlies an age-old fascination with dreams and their mysterious world. But it is only recently that we have gained some understanding of what happens in the brain

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when consciousness is lost and regained, though many unknowns remain. Freud considered dreams as "the royal road to … the unconscious …[1]." Paraphrasing Freud, we might rather say that sleep is the royal road to the conscious.

By consciousness we mean the presence of experience—'what it is like' to see, hear, feel, or have a thought[2]. In this review, we first describe how consciousness changes across different states of sleep and wakefulness, emphasizing the remarkable similarities of dreaming and waking experiences. After briefly reviewing changes in brain activity across the sleep-waking cycle, we discuss how comparing awakenings from dreaming vs. dreamless sleep can inform us about which brain regions are likely parts of the neural substrate of consciousness.

Next, we examine the mechanisms responsible for loss of consciousness during sleep and its return when dreaming. Recent experiments indicate that ongoing states of the neural substrate of consciousness support specific experiences regardless of how they are triggered —endogenously or exogenously. However, consciousness is lost whenever widespread neuronal bistability impairs the causal links within the substrate of consciousness.

We then consider another crucial lesson from dreaming sleep: experience can be present, in all its richness, even though the dreamer is disconnected from the environment This suggests that the substrate of consciousness in its current state is sufficient to specify the quality of experience in a way that is fully intrinsic, regardless of extrinsic inputs and outputs.

Dreams also demonstrate that consciousness can be dissociated from metacognition and cognitive control: when we dream, we can be vividly conscious even though we are not able to reflect, direct our thoughts and actions, and exert volition. Moreover, consciousness can be dissociated from memory acquisition, since we typically don't remember what we dream.

Finally, we consider how internal sources provide a vast and ever-changing repertoire of brain activations that underlie dream scenes and narratives. While the stuff dreams are made of is ultimately an expression of how one's neurons are connected, its fluid recombination into dreams offers a window into the functions of sleep itself.

Experience sampling during sleep: being vs. non-being

From a behavioral standpoint, sleep is characterized by a reduced responsiveness to environmental stimuli, usually associated with immobility and stereotyped postures[3]. This reduced responsiveness is rapidly reversible: behavioral arousability distinguishes sleep from both anesthesia and coma (see Box 1: Sleep and arousability).

Sleep is at once the most dramatic and the most ordinary among the modifications of consciousness. Sometimes, awakening from deep sleep, we realize that, a moment before, we were simply not there. In fact, consciousness could also be defined as "what goes away when we fall into dreamless sleep." Were it not for dreamless sleep, it would be hard to imagine that consciousness is not a given, but something that can come and go, grow and shrink, and that it depends strictly on the way our brain is functioning. And it would be

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difficult to imagine that unconsciousness is truly “non-being”—the absence of everything, ourselves included.

At other times, we awaken to realize instead that, a moment before, we were in the midst of an exciting dream. While in reality we were lying in the dark, our eyes closed, in a small room in the city, we felt fully and convincingly immersed in a realistic seascape, vast and colorful, over which we floated effortlessly. Just as dreamless sleep represents the paradigm of unconsciousness, dreaming sleep provides another fundamental insight concerning consciousness: that “being” does not require doing anything at all—the brain can support experience whether or not we are interacting with the external world.

Assessing the properties of our experiences is at once extraordinarily simple and difficult. It is simple because we just need to report what was going through our mind a moment before (“experience sampling”). It is difficult because the experience is always private and often hard to remember and introspect, whether experience sampling occurs during wakefulness or during sleep.

Examples of the kinds of reports one can obtain from experience sampling during sleep are provided in Fig. 1, in a subject probed through a serial awakening paradigm[4,5]. Report #1, which was obtained towards the morning during rapid eye movement (REM) sleep, is an example of what everybody would recognize as a typical dream: a complex, temporally unfolding episode occurring during sleep that strings together many experiences into a narrative. On another awakening during REM sleep (report #2), the subject reports that she was dreaming, but could not hold the content of the dream in her memory—a so-called “white dream.” Some reports obtained earlier in the night during NREM sleep reveal simpler experiences—often just a single scene—or even just thoughts (reports #3 and 4). Not surprisingly, in the latter case subjects may occasionally assume that they were awake[6]. Finally, another awakening during NREM sleep early in the night produces a report of no experience (report #5). Such “null” reports of “non-being” occur less than a third of the time. If we trust the subject’s words, we have to conclude that her consciousness was absent or at least greatly reduced. Overall, one third of NREM sleep awakenings yield null report, another third yield white dreams, and the rest some remembered content. Almost all REM sleep awakening yield dream reports, although white dreams can also be obtained[7].

After its discovery, REM sleep became synonymous with “dreaming” sleep, because it was only after REM sleep awakenings that subjects regularly reported they “had a dream,” in the sense of vivid scenes unfolding along an intricate narrative[8]. However, when subjects were asked, in subsequent studies, about any content that might have gone through their mind, it became clear that they experienced scenes or thoughts through much of NREM sleep. Moreover, in terms of contents, the properties of NREM and REM sleep dreaming tend to fall along a continuum[9,10]. At most, NREM reports have fewer bizarre elements and aggressive social interactions (but more friendly interactions). This led support to the “one-generator” model, according to which dreams are produced by the brain in a way that is only quantitatively different between NREM and REM sleep[11–14] (see[15] for a discussion).

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Despite these similarities, a striking feature of REM dreams remains their length and narrative organization. For example, after just 5 minutes of REM sleep, raters could recognize an average of 5-6 concatenated "dramatic units" vs. ~1 unit after an arbitrary duration of NREM stage 2 sleep[16] (but see[17]). Moreover, the typical duration of dream reports (estimated based on word count) is 3 to 7 times longer for REM dreams in both home and laboratory studies[18,19]. In other words, typical REM dreams are story-like—much like a play. While the story-like organization of REM dreams is usually evident to the dreamer, it is more difficult to quantify based on transcriptions available to third-party raters (for example, by scoring cause-effect-cause sequences[20] or word linkages with graph-theoretical tools[21]).

A natural question concerns the trustworthiness of dream reports, since what is reported depends not only on what was actually experienced, but also on what was committed to and retrieved from memory[22]. For example, it has been claimed that during sleep we are always unconscious and dream reports are confabulations upon awakenings[23,24]. However, there are good epistemic reasons to take dreams at face value and trust dream reports, as long as they are collected under near-ideal conditions and immediately after awakening[25]. Moreover, we now have empirical evidence that the occurrence, duration, and content of a dream can be predicted based on brain activity preceding the awakenings[7]. Conversely, it has been argued that consciousness may never vanish during sleep, but merely be forgotten. However, careful questioning of experienced subjects, including long-term meditators, indicate that "null" reports (no experience whatsoever) are quite different from "blank" reports (a sense of presence with no content) and from "white" dreams (a sense of having dreamt but forgotten).

Similarities between sleep and waking consciousness

It is instructive to compare reports obtained upon awakening from sleep with those obtained through experience sampling when a subject is already awake. For example, report #6 (Fig. 1) was obtained by interrupting the same subject while waiting for the start of the experiment. Reports #7 and 8 (the latter a case of "blank mind") were obtained when interrupting her while she was being set up for the EEG recording. Reports #9 and 10 were obtained when she was transitioning from wakefulness to sleep (see Box 2: Sleep onset experiences).

These examples show that there are great variations in the kinds of experiences that occur across behavioral states, from experiences tied to a task to free-roaming ones, from vividly perceptual ones to thought-like and abstract ones, from isolated images or thoughts to evolving narratives. They also show that these kinds of experiences can occur partly independent of behavioral state. For example, ~20% of reports obtained from subjects who were awake but not engaged in a task can be qualitatively indistinguishable from typical dream reports[26]. Studies of mind-wandering have confirmed that subjects often report being lost in thought and unaware of the environment[27]. In fact, some consider dreaming as an intensified form of mind-wandering[28]. Conversely, states of lucid dreaming, in which subjects are aware that they are dreaming and may regain voluntary control of their thoughts and images, share features of awake, goal-directed consciousness (see below). Thus, a main conclusion from decades of studies using experience sampling is that conscious states

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transcend the boundaries of behavioral states. Moreover, dimensions such as connectedness to the environment, voluntary control, metacognition, thought, and perceptual vividness can be present to various degrees in different states of consciousness, whether awake or asleep.

The analysis of specific contents of experience also shows that what one can be conscious of is similar in sleep and wakefulness. Thus, waking consciousness is dominated by the visual modality, closely followed by the auditory modality, both bound to a spatial framework. Similarly, experiences occurring during sleep are almost always visual in nature, often auditory, and usually spatial[29]. Other modalities seem to be less represented in dreams, but that may be partly an artifact of what we can more easily remember or report Feelings of floating, instead, are more frequent in dreams. The categories that are the stuff of dreams are the same as those that constitute the fabric of wakefulness—objects, animals, people, faces, and places. Dream experiences are not necessarily all vivid and perceptual—there are also faint ideas, just as in wakefulness, and we can have all sorts of thoughts while we dream. Hearing speech or conversation is also extremely frequent, and speech patterns are as grammatically correct as in waking life. Emotions are present, both positive and negative, and so are feelings of agency. Altogether, it appears that, if some quality of consciousness can be experienced during wakefulness, it can also be experienced during sleep, and the other way around. In fact, subjects who have lost the ability to experience some contents during wakefulness, such as faces, also lack that ability when dreaming[30]. If people can still experience visual images, they have visual dreams, otherwise not[31].

Finally, the issues one thinks about or explores during wakefulness—a person's daytime concerns and preoccupations—are very much the same that are explored during dreaming sleep. This is known as the "continuity" between the waking and the dreaming self[32,33]. In short, the ingredients that compose dreaming experiences are the same as those that compose waking life—what differs is how they are triggered and combined.

Brain activity during sleep

Given the above conclusion, what can we learn from the study of neural activity across sleep and wakefulness and its relationship to consciousness?

Sleep is traditionally subdivided into stages based on behavior and on physiological variables used in polysomnography, including EEG, muscle tone, and eye movements. In mammals, the main subdivision is between sleep without or with rapid eye movements (NREM and REM sleep). NREM sleep is characterized by EEG slow waves and spindles. Within NREM sleep, one distinguishes between the brief falling asleep stage (N1) accompanied by rapid changes in the EEG, light sleep (N2), with spindles and few slow waves, and deep sleep (N3) with more slow waves. REM sleep is further characterized by an activated EEG (low voltage, fast activity) and by sawtooth waves. In humans, cycles of NREM and REM sleep repeat 4-5 times per night, with REM sleep making up for ~20% of sleep time. In the course of sleep, the amplitude of slow waves decreases progressively, while the duration of REM sleep episodes increases[34].

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NREM sleep

Unit recordings, both extra- and intra-cellular, have revealed what underlies the main signatures of sleep (Fig. 2). Sleep slow waves—the most widespread feature of NREM sleep—are the EEG correlate of synchronous fluctuations in the membrane potential of cortical neurons, which alternate between a depolarized phase (up-state), during which neural activity is nearly as high as during wakefulness (ON-period), followed every second or so by a hyperpolarized phase (down-state) of silence (OFF-period), lasting hundreds of milliseconds[35,36]. During natural sleep, OFF-periods are mostly local and do not engage the entire cortical mantle[37]. OFF-periods can also occur during wakefulness under conditions of sleep deprivation[38,39]. The thalamus also helps synchronizing cortical slow waves[40–42], and the claustrum may play a similar role[43]. Moreover, slow waves invade the basal ganglia[44] and the cerebellum[45].

Spindles are generated by inhibitory cells in the reticular nucleus of the thalamus through the interplay between their intrinsic properties and interactions with bursting thalamocortical relay neurons[35]. Spindles provide bouts of rhythmicity to target neurons in cortex and elsewhere, and they can be variously coupled to the larger fluctuations in membrane potential that characterize slow waves and sharp-wave ripples[46].

Throughout NREM sleep, the hippocampus is invaded by cortical volleys that often trigger sharp wave-ripple complexes, which are relied back to cortex (the dentate gyrus generates its own kind of “dentate spikes”[47].) Sharp waves, generated by synchronous input volleys from CA3 to the apical dendrites of CA1 neurons, are typically followed by a brief (<100 milliseconds) oscillatory (100-200 Hz) ripple. During the ripple, neurons spike sequentially (ripple sequences) in a way that reflects the strength and directionality of the underlying connectivity[48,49]. In rodents, these ripple sequences often represent coherent spatial trajectories[50]. Hippocampal sharp waves also occur during quiet wakefulness. In rodents this can happen when the animal is engaged in behaviors such as eating, drinking, or grooming[48,49]. In humans, sharp waves are produced, for example, during memory recall[51,52].

REM sleep

During REM sleep cortical neurons show a sustained depolarization, similar to wakefulness, uninterrupted by OFF-periods[36]. This tonic firing underlies its low voltage, fast activity EEG, punctuated by phasic bursts of sawtooth waves, which usually accompany rapid eye movements. In many regions of the rodent brain, this sustained, tonic firing is superimposed on membrane potential fluctuations in the theta range. The theta rhythm is especially evident in ventromedial prefrontal cortex, hippocampal formation, retrosplenial cortex, and anterior thalamic nuclei, as well as in hypothalamic and brainstem nuclei[53]. The human equivalent of rodent theta rhythm may be a slower rhythm (1–2.5 Hz) that can be recorded in homologous regions[54–56]. The rhythm is expected to facilitate interactions among distant brain regions. In the hippocampus, neuronal firing during theta activity is arranged in theta sequences: at every cycle of the rhythm, neurons fire in fast sequences (<100 msec) reflecting the strength and directionality of the underlying connectivity. In awake rodents, theta activity occurs during exploratory behaviors, and a single theta sequence involves a subset of neurons

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activated within ~2 sec[57]. Theta activity during REM sleep appears to be more regular and extended than during wakefulness, though there are some differences[58], including slightly slower theta sequences[59].

REM sleep is also characterized by phasic phenomena, such as the rapid eye movements that gave it its name, which tend to occur every minute or so[60]. In the brainstem, specific populations of neurons become activated at the transition between NREM and REM sleep, firing bursts of action potentials that invade tectum, thalamus, and several cortical areas, triggering ponto-geniculo-occipital (PGO) waves. Individual PGO waves anticipate REM sleep by several minutes, after which they often occur in clusters and are frequently accompanied by REMs. This has led to the suggestion that many dream reports obtained from polysomno-graphically defined NREM sleep may instead be elicited from *covert* REM sleep[9]. In humans, sawtooth waves, likely related to PGO waves, also occur in clusters and anticipate the transition from NREM to REM sleep by several minutes. They are prominent in retrosplenial cortex and mesial temporal lobe areas, insula, frontal operculum, and sensorimotor cortex, and are accompanied by intense gamma activity[61,62]. In rodents phasic activity is prominent in retrosplenial cortex at the transition from NREM to REM sleep as well as during REM sleep[63,64].

Local sleep and dissociated states

Over the past several years, we have learned that sleep is not a monolithic, global state (nor, for that matter, is wakefulness). Local field potential recordings have enriched our understanding of neural activity during sleep based on scalp EEG, revealing that sleep has both global and local, synchronous and asynchronous features[37,65,66]. For example, the thalamus can “deactivate” several minutes before the cerebral cortex[67,68]. Slow waves occur in different cortical areas at different times, like bubbles in boiling water ([37], see also [69]). Spindles, too, are local[37], and can appear in the hippocampus up to 30 minutes before the neocortex[70]. Transitions between NREM and REM sleep can also vary across areas[71]. Sawtooth waves anticipate the overt transition into REM sleep and are localized to a subset of brain regions[61,62,71]. Regional dissociations are also found within cortex. For example, primary areas show local slow waves during REM sleep[72–74]. Furthermore, there can be micro- and partial arousals throughout sleep, during which some of the typical activity patterns of wakefulness are reinstated, but briefly and in some regions and not others[65,75]. Even full arousals are not synchronous across brain areas[76–78], which may underlie the so-called sleep inertia[79]. In fact, local EEG slowing can occur even during wakefulness if subjects have been sleep deprived[38,80].

As we will see, these local aspects of brain activity during sleep and wakefulness must be considered whenever one tries to establish the neural correlates of consciousness.

The neural correlates of consciousness in sleep: within-state paradigms

The variety of neural activity patterns across sleep and wakefulness, coupled with the variety of subjective states revealed by experience sampling, lends itself naturally to exploring the neural correlates of consciousness. However, simply comparing wakefulness with NREM sleep, or NREM with REM sleep, cannot reveal the neural correlates of consciousness.

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This is because, as demonstrated by serial awakenings, consciousness is present in all three states[5]. Furthermore, many aspects of brain physiology change between wakefulness, NREM, and REM sleep, above and beyond any changes associated with being conscious or not In fact, the very observation that both dreaming and dreamless sleep could occur with a "synchronized" EEG (NREM sleep) had led some to doubt any correspondence between brain activity and consciousness[81].

On the other hand, serial awakenings clearly demonstrated that consciousness can be present or absent *within* the same sleep stage. This provides the opportunity to distill the neural correlates of being conscious at a time when subjects are not behaving, are disconnected from the environment, and brain activity remains relatively stable. Furthermore, high-density EEG recordings can detect regional differences in key features of sleep, such as slow waves and spindles.

The results of a within-state, no-task paradigm using serial awakenings (>1000) and high-density EEG are summarized in Fig. 3A[7]. When subjects awakened from NREM sleep (stage N2) reported having dreamt, slow wave activity (power in the 1 to 4 Hz range) in the EEG before the awakening (20 seconds) was lower than when they reported no experiences, but only over a set of primarily posterior-central cortical regions. The results were similar when white dreams were compared to dreamless sleep, suggesting that the reduction in slow waves in posterior cortical areas correlates with the presence of consciousness rather than with the ability to recall a dream. A similar but broader set of areas showed increased high-frequency activities during dreaming vs. dreamless sleep, in line with the observation that slow waves are associated with a reduction in high-frequency power[82].

Very few awakenings from REM sleep yield null reports. Despite the reduced statistical power, a within-REM sleep comparison also revealed that dreaming REM sleep was associated with lower slow wave activity in posterior cortical areas. Thus, within-state comparisons across two very different sleep stages—NREM and REM—yielded consistent results: consciousness is lost when slow wave activity increases primarily in posterior cortical areas. (The results were similar between experienced subjects who were awakened many times over a week, and naive subjects who were awakened a few times over one night).

A follow-up study focusing on slow waves[83] found that dreaming was most likely if slow waves over posterior-central areas were sparse, small, with a shallow slope, and had many intra-wave negative peaks (type II waves[84]). Dreaming was also associated with the occurrence of fast spindles over posterior-central brain regions. Spindles tend to be faster when the membrane potential of thalamic neurons is more depolarized or becoming more depolarized[85], as in the transition to the ON-period[84,86]. By contrast, dreamless sleep was associated with slow waves that were frequent, large, with a steep slope, had few intra-wave negative peaks, and were accompanied by slower spindles and substantially reduced gamma activity. Another study using serial awakenings confirmed that dreaming during NREM sleep (N2) was associated with diminished low-frequency activity (< 1Hz) and higher power above 4Hz[87]. Yet another study found a reduction of slow wave activity over left frontal

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and temporo-parietal areas during naps with dream recall although, with 28 electrodes, localization was less precise[88].

Altogether, these experiments demonstrate that dreaming is likely if slow wave activity is low over posterior-central cortical regions, regardless of the sleep stage. This is almost always the case during REM sleep, when slow waves are small and localized[73], and increasingly so during NREM sleep late in the night. Instead, dreamless sleep is most likely during slow wave sleep early in the night, when slow waves are frequent, large, and often invade posterior-central cortex. Notably, dream reports virtually vanish after sleep deprivation, when recovery sleep is characterized by markedly increased slow wave activity[89].

Dreaming vs. remembering the dream

While both remembered dreams and white dreams were associated with reduced slow wave activity over posterior-central cortical regions, their contrast revealed greater high-frequency activity around mid-cingulate cortex for remembered dreams. Thus, encoding a transient trace of the dream, or being able to recall it, seems to require additional circuits beyond those necessary to experience it. Alternatively, white dreams may simply be less vivid, closer to the threshold of reportability, just like near-threshold stimuli or “glimpses” in the case of perception[90]. Further work showed that remembered NREM dreams were often preceded by isolated, high amplitude slow waves in fronto-central regions followed by a local increase in high-frequency activity[83]. These type I slow waves[84], which include K-complexes, are likely triggered via the activation of brainstem arousal systems by sensory stimuli of various modalities[91]. They may promote partial, intermittent awakenings and the encoding of memory traces and thus permit the recall of some dream content[92].

Neural correlates of dream contents

Within-state paradigms can also be applied to specific dream contents[7], lending further confidence to the conclusion that posterior-central cortical areas are the substrate of dreaming consciousness. Within REM sleep dreams, reports rated by the subjects as primarily sensory in quality were associated with increased gamma activity over posterior, sensory cortices. Instead, thought-like content was negatively correlated with slow wave activity in mid-cingulate regions across REM sleep, NREM sleep, and experience sampling during wakefulness[93]. REM sleep dreams containing faces—almost half of all—were associated with higher gamma power over fusiform face area[7] (Fig. 3B). Dreams containing speech, compared with those without, showed a localized increase in high-frequency power over left temporo-parietal cortex, as expected from neuropsychological data. Similarly, dreams tied to a spatial setting showed increased gamma power over right posterior parietal cortex. Dreams involving body movements were associated with activation of the posterior part of the right superior temporal sulcus, an area typically activated when awake subject watch people or animals moving[7], while dreams with frightening content activate the midcingulate gyrus and the insula, as fear does in wakefulness[94]. Importantly, the localized increase in gamma activity associated with specific contents, such as faces (Fig. 3B), was most consistent for the 2 seconds before awakening and became less evident when considering longer intervals[7]. Because subjects had been instructed to focus on the last scene

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experienced in the dream, this suggests that the dream report captured the content of what was just experienced with good temporal accuracy.

In summary, within-state paradigms demonstrate that the neural correlate of being conscious are primarily located in posterior-central cortical regions. It should be noted, however, that average within-state comparisons based on dream reports inevitably underrepresent the richness of content of individual dreams. The areas that show reduced slow wave activity are likely those that are activated in most dreams, reflecting typical contents, such as visual imagery. Neural correlates of contents that may have occurred in just a single dream, or just a single subject, would not be readily apparent This is likely the case for modalities that are rarely experienced in dreams, such as taste or smell, and for some abstract contents. Also, while within-sleep paradigms avoid contamination by task-related activities, they may be sensitive to other neural processes, such as unconscious processes that may be implicated in the generation of dreams.

Neural correlates of consciousness: beyond sleep

The results of contrasts between consciousness and unconsciousness performed within the same sleep stage (NREM or REM), which point to posterior-central regions, are consistent with recent results obtained during “anesthesia dreaming”[95]. As with sleep, loss of consciousness is associated with increased slow wave activity in posterior cortex during propofol anesthesia[96] and dexmedetomidine infusions[97].

Lesion, stimulation, and recording experiments also highlight the importance of posterior cortical areas[98]. Subjects with extensive bilateral traumatic damage posterior to the central sulcus virtually never recover from a state of unresponsive wakefulness[99]. Also, the extent of damage to posterior cortical areas after cardiac arrest predicts lack of recovery better than overall brain damage[100,101]. Furthermore, damage to specific regions of posterior cortex can permanently abolish specific contents of experience. For example, subjects with lesions or developmental abnormalities affecting the fusiform face area are not just unable to recognize faces (prosopagnosia), but do not realize that they are missing an aspect of experience others enjoy (anosognosia). In fact, they can neither imagine a face, nor dream of one[30]. Along the same lines, posterior-central cortical areas are also likely to trigger specific contents of experience when microstimulated electrically in epileptic subjects[102,103]. Evoked feelings, which are location-specific, include lights, sounds, touch, a sense of falling or sliding, and so on. For example, stimulation of the right fusiform face areas produces distorted or illusory perceptions of faces[104,105]. Finally, regions activated by dreaming of specific features, such as faces, are also activated by appropriate sensory stimuli during wakefulness, and their activation persists for as long as the stimulus is perceived, which is not true for other cortical areas[106]. In fact, the most reliable, task-independent EEG correlate of the conscious perception of visual, auditory, and somatosensory stimuli is an early negativity in the EEG evoked response (around 200 milliseconds or so) over the corresponding posterior sensory cortices[107].

On the other hand, large portions of prefrontal cortex, which fail to show EEG differences between dreaming and dreamless sleep, do not seem to contribute directly to consciousness[98]. Lesion studies demonstrate that prefrontal regions are instead essential

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for cognitive functions such as attention, working memory, task initiation, monitoring, and switching, as well as for metacognition[108]. Moreover, microstimulation of prefrontal cortex usually yields no experiential effects, with the exception of some portions of orbitofrontal and ventromedial prefrontal cortex[102].

The evidence suggesting that posterior-central cortical areas are the likely substrate of consciousness in our brain is in line with the theoretical predictions of integrated information theory (IIT[2,109–111]). According to the theory, the substrate of experience should be a dense lattice of neural elements that can interact effectively over the time scale of experience. Posterior-central cortical regions conform to the anatomical requirements of a tight lattice[112], unlike other brain regions[113]. Searching fMRI data (from more than 1000 subjects) for cortical regions more densely functionally connected among themselves than to other regions areas[114] (see also[115]) also reveals a "functionally rich club" that includes precuneus, posterior cingulate, and midcingulate cortices but not prefrontal areas. The above evidence is also in line with the proposal that recurrent interactions in posterior-central cortical areas are sufficient for sustaining consciousness[116,117]. On the other hand, it is not consistent with the idea that prefrontal cortex is necessary for consciousness.

Loss of consciousness during sleep: neuronal bistability and the breakdown of causal links

An association between consciousness and "brain activation," understood as low-voltage fast activity in the absence of slow waves, had been postulated from the early days of EEG recordings[118]. In fact, it was explicitly proposed that dreaming consciousness would be a function of brain activation[11]. But the occurrence of dreaming during NREM sleep, which is defined by the presence of slow waves, presented a paradox. The answer to this paradox is that slow waves can be local: dreaming happens when posterior-central cortical areas are relatively activated, even though slow waves may dominate in more anterior areas (hence the EEG is scored as NREM sleep)[7]. But why would the occurrence of slow waves over posterior-central cortex lead to loss of consciousness?

Neuronal OFF-periods and bistability

As shown in Fig. 2, neuronal firing during the up-state of sleep slow waves is not dissimilar from that observed during wakefulness and is usually accompanied by gamma activity in the local field potential[119]. However, sleep up-states are invariably interrupted by the occurrence of down-states and the collapse of gamma activity, which makes cortical circuits "bistable".

The synchronous occurrence of OFF-periods involving a large number of neurons is due at least in part to the broad inhibitory action of somatostatin expressing neurons, among them Martinotti cells[120–123]. In the case of pyramidal neurons, a major target of this inhibition is the apical dendrite through both pre- and post-synaptic $GABA_B$ mechanisms whose action can last for a few hundreds of milliseconds. Adaptation mechanisms, such as a surge in activity-dependent potassium currents, are also involved in bistability[124,125].

As shown by experiments in humans, animals, and in silico, the degree of bistability can be estimated by the features of sleep slow waves[126–128]. The number of slow waves reflects

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the propensity of cortical neurons to enter the down-state, while their amplitude reflects the number of cortical neurons that do so together. The descending slope of the wave reflects how synchronously neurons enter the down-state, and the ascending slope how synchronously they enter the up-state. Multiple negative peaks within a slow wave indicates that down-states are not well synchronized across distant neuronal populations. From this perspective, the high-amplitude, steep slow waves that sweep through the cortex more than 10 times a minute in early NREM sleep (N3), when consciousness often vanishes, indicate high bistability. Bistability attenuates during NREM sleep later in the night (N2), when slow waves become less frequent, more asynchronous, and localized mostly in the front, and dreaming is common, even though often fragmentary. Bistability is even lower at sleep onset (N1), when posterior cortical areas may remain activated for longer, and subjects are almost invariably conscious. Finally, bistability is altogether absent during most of REM sleep, when subjects are not only conscious, but tend to enjoy prolonged, narrative dreams.

Breakdown of causal links during dreamless sleep

Why would bistability interfere with consciousness, even though down states only last for fractions of a second, and neurons continue to fire throughout sleep? Theoretical considerations concerning the physical requirements for consciousness predict that down-states should lead to unconsciousness if they lead to the breakdown of causal links within the substrate of consciousness[109,111]. This prediction was tested in subjects with epilepsy implanted with intracranial electrodes for clinical evaluations[82] (Fig. 4A). In these subjects, electrical stimulation during wakefulness triggered a chain of deterministic phase-locked activations at other cortical electrodes. However, during deep NREM sleep, the same stimulation triggered a slow wave associated with a suppression of EEG power above 20 Hz, after which phase locking was lost. This was the case even though cortical activity had resumed to levels similar to those observed in wakefulness. Recent studies in rodents produced similar results, further demonstrating that the loss of phase locking is associated with neuronal OFF-periods[129] (Fig. 4B). These findings suggest that neuronal bistability—the tendency of cortical neurons to fall into an OFF-period after a transient activation—breaks the causal links among connected populations. From a theoretical perspective, this breakdown of causal links disintegrates the substrate of consciousness into small modules (loss of integration). Moreover, it collapses its repertoire of available states because different neural activity patterns during ON-periods do not lead to differentiated effects, but always to the same state—an OFF-period (loss of information[130]).

These results also provide a mechanistic understanding of previous work that employed transcranial magnetic stimulation (TMS) to assess information and integration across wakefulness and sleep[131]. A key advantage of using TMS is that, unlike sensory stimulation, direct cortical stimulation does not activate the reticular formation and bypasses the thalamic gate (see below). Thus, it directly probes the ability of cortical areas to interact. As shown in Fig. 4C, TMS during wakefulness induces a sustained response characterized by a sequence of time-locked, high-frequency oscillations (20-35 Hz) for the first 100 milliseconds, followed by several slower components (8-12 Hz) that persists until 300 milliseconds. Source modeling shows that the initial local response to TMS is followed by rapidly changing configurations of activations, likely mediated at first by cortico-thalamo-cortical

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loops and then by long-range ipsilateral and transcallosal connections. Not surprisingly, the pattern of activation switches to different cortical areas depending on the connectivity of the site that is stimulated.

At sleep onset, the initial TMS-evoked response increases in amplitude but becomes progressively shorter in duration because later waves are dampened (Fig. 4C). During NREM sleep proper, and especially during N3, the brain response to TMS changes markedly. The initial response doubles in amplitude and lasts longer, after which no further TMS-locked activity can be detected, and all TMS-evoked activity ceases by 150 milliseconds. This response, which lacks high-frequency components, remains localized and does not propagate to connected brain regions. When applied to a median centroparietal region, TMS pulses can trigger high-amplitude slow waves that resemble spontaneous type I waves[84] and travel through much of the cortex[132]. The stereotypical negative peak evoked by the TMS pulse, presumably corresponding to a widespread cortical down-state, suggests a loss of differentiation of neural responses. Importantly, during NREM sleep stereotypical responses can be induced even when the surface EEG does not display spontaneous slow waves for several seconds, unmasking the underlying bistability of the corticothalamic system[125].

A subsequent study showed that, within NREM sleep, the response to TMS differed according to the subjects' report upon awakening[133]. Reports of no dreaming were associated with a larger negative evoked response and shorter phase-locking compared with reports of dreaming. Furthermore, the amplitude of the negative peak, presumably reflecting the degree of neuronal bistability, was inversely correlated with the length of the dream report (word count). As we have seen, the main difference between dreamless and dreaming NREM sleep is the presence of higher slow wave activity over posterior-central cortical areas. It remains to be established, using perturbational approaches, whether dreaming NREM sleep is characterized by bistability mainly in prefrontal areas, but not in posterior-central areas.

Finally, during REM sleep, especially late in the night, when dreams become long and vivid, the responses to TMS recover and come to resemble more closely those observed during wakefulness (Fig. 4C): evoked patterns of activity become more complex and spatially differentiated, although some late components may be reduced compared to wakefulness[134]. The difference in brain responses to TMS or electrical stimulation between states of consciousness and unconsciousness can be quantified by a numerical index—the perturbational complexity index (PCI)[135]. In rodents, as in humans, PCI values are high in wakefulness and REM sleep and low in NREM sleep[129] as well as in anesthesia[129,136,137].

Breakdown of causal links and loss of consciousness: beyond dreamless sleep

Altogether, these results indicate that loss of consciousness is associated with a breakdown of causal links within cortical (and cortico-thalamic) circuits owing to bistability. These findings have been further corroborated by studies employing TMS-EEG in general anesthesia and disorders of consciousness. Thus, during midazolam anesthesia, TMS induces an EEG response that is both shortlived and local[138]. Similar results were obtained with different anesthetics, including propofol (which also produced a local response

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resembling a type II slow wave) and xenon (which produced a large, stereotypical, low complexity response resembling a type I slow wave). A noteworthy exception is ketamine anesthesia, during which TMS triggered a longer-lasting, complex spatiotemporal activation of the EEG similar to what is observed during wakefulness and especially REM sleep. While participants emerging from the other anesthetics did not report any experience, after ketamine they reported long, vivid dreams, unrelated to the hospital environment[139]. Since these initial studies, PCI has been validated in many experimental and clinical conditions[140] and is currently the most sensitive and specific measure to detect covert consciousness in clinical settings[141].

The inverse relationship between bistability and consciousness is in line with the theoretical predictions of integrated information theory (IIT[2,109–111]). According to the theory, given an appropriate anatomical substrate (such as brain areas with a dense, lattice-like connectivity), what matters is the potential for causal interactions—i.e. causal "competence"—rather than the actual carrying out of some specific function, computation, or process. When bistability ensues, causal competence is lost, and with it consciousness, regardless of the current activity pattern.

In addition to bistability, another potential mechanism underlying loss of consciousness, initially proposed in the context of anesthesia, is the breakdown of dendritic integration in thick-tufted layer 5 pyramidal neurons[142]. The apical tuft receives input primarily through feedback from higher cortical areas, whereas the basal dendrites receive primarily feedforward sensory input. During wakefulness, consciousness would be supported through the coupling of basal and apical inputs, whenever a calcium spike generated by the apical tuft coincides with somatic sodium spikes. This would lead to the transition from tonic firing to high-frequency bursting, which would then close cortico-cortical and cortico-thalamic loops and support consciousness. Indeed, in a mouse model this bursting is prevented by anesthesia[143]. Similar mechanisms might explain the loss of consciousness in dreamless sleep. Conversely, dreaming during REM sleep might be supported by an "apical drive," when neuromodulatory changes, especially the high levels of acetylcholine, would allow the apical tuft to drive the firing of the soma in the absence of sensory inputs[144].

Dreaming: consciousness without connectedness, reflection, and memory acquisition

The previous sections have highlighted that consciousness is present, regardless of behavioral state, whenever its neural substrate is not in a state of bistability, which disrupts causal links inside it But dreaming consciousness reveals another fundamental fact: that the substrate of consciousness fully specifies how an experience feels and does so intrinsically, regardless of interactions with the environment. Moreover, dreams indicate that consciousness can be dissociated from reflection as well as from memory acquisition.

Consciousness disconnected

The most striking behavioral consequence of falling asleep is a progressive disconnection from the outside world, even though we may continue to be conscious. For example, in an

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early study, subjects slept with their eyes taped open and the pupil dilated, but the bright visual stimuli they were shown did not influence the content of the dream, which followed its own visual narrative[145]. Even erections, which accompany 90% of REM sleep episodes, are rarely associated with erotic dreams. Disconnection is sleep's defining feature and, given its inherent dangerousness, it is likely essential to its function[3].

One should distinguish between behavioral, neurophysiological, and phenomenal disconnection. *Behavioral disconnection* is manifested because a sleeper fails to appropriately respond to stimuli. This is typically evaluated by the "arousal threshold"—the stimulus intensity that will arouse—that is, awaken—a subject The arousal threshold from slow wave sleep (N3) and REM sleep can be very high: people, especially small children, can fall asleep in a noisy environment and in uncomfortable positions. However, in contrast to a person under anesthesia or in a coma, a sleeping person can always be awakened if stimuli are strong enough or meaningful, such as the sound of one's name or the wailing of a baby (see Box 1: Sleep and arousability).

Neurophysiological disconnection means that stimuli may not percolate within the brain as far as they typically do during wakefulness. Changes in the brain's response to sensory stimuli are certainly not surprising given the marked changes in neuromodulation that characterize NREM and REM sleep, but how stimuli may be "gated" remains unclear[146].

One possibility is a *brainstem gate*. During wakefulness, sensory stimuli activate both primary cortical areas and brainstem cell groups, including glutamatergic cells in the mesencephalic reticular formation and noradrenergic locus coeruleus cells in the dorsal pons. The latter can trigger a "reset" of collicular, thalamic, and cortical activity[147] that may underlie an attentional resetting[148]. The P300 wave, typically evoked by surprising stimuli, may represent the EEG correlate of such resetting[147]. However, during NREM sleep the P300 wave is abolished and during REM sleep it can only be evoked by very rare and loud stimuli[149], suggesting that many stimuli may not get through to the cortex, or fail to produce sustained responses, if they fail to activate the locus coeruleus and other brainstem circuits. In fact, during REM sleep the locus coeruleus is actively inhibited[150], along with other brainstem circuits, which may prevent attentional resetting and cause stimuli to be overlooked.

Another possibility is that sleep enforces a *thalamic gate* through hyperpolarization[151]. However, it is clear that such a gate is at best partial (see[152] for a recent review). During NREM sleep, sensory stimuli in various modalities can still elicit evoked potentials from the cerebral cortex[153], and neuroimaging studies show that primary cortical areas can still be activated[154,155]. During REM sleep and at the transition between NREM and REM sleep, however, spindling in first-order thalamic nuclei is almost continuous, and may effectively work as a "guardian" of sleep[46].

The bistability of cortical circuits during NREM sleep may enforce a *cortical gate*[156] due to the breakdown of intracortical links during OFF-periods. Indeed, while sensory stimuli arriving at the cortex during the transition to the up-state of the slow oscillation elicit evoked responses comparable to those seen in wakefulness, the response is much reduced at

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the transition to the down-state, and later components may disappear altogether (reviewed in[146]). Neuroimaging studies also show that, while simple acoustic stimuli can activate primary auditory cortex during NREM sleep, they do not reach much beyond it[154]. As already mentioned, during REM sleep slow waves are present in primary sensory and motor areas in both rodents (mostly in layer 3 and 4[72]) and humans[62,73] (consistent with the reduction of activity in primary visual cortex[157]). Thus, bistability in primary areas may partially explain sensory disconnection during REM sleep. One wonders whether bypassing thalamic or cortical gates through the direct activation of higher-level cortical areas, say through TMS, would instead readily elicit experiential contents.

There is also evidence that sensory stimuli can occasionally reach deep within the cortex, up to motor areas, in both NREM and REM sleep. For example, during tonic REM sleep meaningful speech can be decoded from EEG activity better than meaningless speech, suggesting that sensory input can reach high-level areas of the brain[158]. In another study, subjects were able to signal words vs. pseudowords by frowning or smiling while asleep in either N2 and REM sleep, and their EEG showed signs of activation[159]. These findings are in line with the evidence, mentioned earlier, that states of sleep and wakefulness can be locally heterogenous and asynchronous. What is unclear is whether subjects consciously heard the stimuli, since they could not recall.

Although consciousness is present through much of sleep, the contents and narrative of dreams are largely independent from the environment This *phenomenal disconnection*—a paramount feature of sleep—occurs even when sensory stimuli can trigger activations within the neural substrate of consciousness. This is especially puzzling in the case of REM sleep, given that we are usually vividly conscious and most of the cortex shows no signs of bistability.

One possibility is a disruption of top-down attentional mechanisms[160]. As we have seen, several prefrontal and parietal cortical areas are deactivated in REM sleep, including areas important for directing and sustaining top-down attention to sensory cortices. Even if stimuli were to reach the substrate of consciousness, they may be left effectively unattended, similar to inattentional blindness or deafness[161]. Some evidence is provided by unit and local field potential recordings in subjects with epilepsy presented with various auditory stimuli[162]. Compared to wakefulness, spiking and high-gamma responses to sounds in lateral temporal cortex were only mildly reduced in NREM and REM sleep (mostly late responses in higher cortical areas). However, sleep strongly affected auditory-induced alpha–beta (10–30 Hz) desynchronization, which might reflect a lack of top-down attention, perhaps similar to what happens in neglect[163]. The interplay between extrinsically oriented, task-relevant networks, and intrinsically driven, default-mode networks, may also contribute to a redirection of attention[28].

Another possibility is that the activation triggered by exogenous stimuli during REM sleep may be incongruent with the activation triggered endogenously by the dream. Specifically, similar to what may happen during binocular rivalry[164], the causes and effects of dream states would be incompatible with those specified by the stimulus, and therefore remain unconscious, in line with theoretical notions[109].

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Finally, stimulus-triggered activations that reach the neural substrate of consciousness may be incorporated into the ongoing dream experience and reinterpreted without awakening the dreamer[165]. There is indeed evidence that certain stimuli, such as pressure on the limbs[166] or meaningful words[167,168] can become part of the dream narrative, though this is the exception rather than the rule. The external perturbation is typically interpreted in a way that fits the nature of the dream, rather than that of the stimulus[169,170]. For example, a sudden external sound may become a sudden movement in the dream. Moreover, even when a dreamt sequence seems purposefully crafted retrospectively to make sense of a stimulus, a more plausible explanation is that a stimulus only gets incorporated if it fits the preceding dream. The classic example is Maury's guillotine dream, quoted by Freud: *"…after a number of incidents which were not retained in his memory, he was condemned, and led to the place of execution surrounded by an immense mob. He climbed on to the scaffold and was bound to the plank by the executioner. It was tipped up. The blade of the guillotine fell. He felt his head being separated from his body, woke up in extreme anxiety—and found that the top of the bed had fallen down and had struck his cervical vertebrae just in the way in which the blade of the guillotine would actually have struck them*"[1, p. 59].

In summary, disconnected or partially disconnected consciousness is the hallmark of dreams and what distinguishes them from perceptions. From the conventional, extrinsic perspective, perception is considered as a representation of the environment obtained by "processing information" or "making inferences" about it that somehow becomes conscious. Dreams are then seen as "immersive simulations" of the world as perceived, where the brain would take on the additional burden of supplying the information to be processed, substituting for the environment Instead, from the intrinsic perspective of a conscious subject[109], dreams show that the substrate of consciousness, largely disconnected from the environment, fully specifies the quality and meaning of an experience through its intrinsic causal powers[109,171]. From this intrinsic perspective, the difference between waking and dreaming experiences is primarily a matter of how they are triggered. Waking perception can then be thought of as a dream, triggered exogenously, that matches causal processes in the world[171].

We will not review the output side of disconnection during sleep, except for noting the well-characterized "brainstem gate" responsible for REM sleep atonia (see below[150,172,173]). This ensures that dreamt behaviors are not translated into actual behaviors, as may happen under pathological conditions[174]. Like for disconnection on the input side, higher-level gates may also contribute, including the deactivation or inhibition of cortico-basal ganglia-thalamocortical circuits and functional disconnections within cortex, as discussed below.

Consciousness unreflective

When we are awake, we usually reflect and exert cognitive control: we regularly stop to think about what is happening, select goals, initiate actions, and actively switch tasks; we control attention and employ working memory to think, manipulate concepts in our mind, remember, and imagine. By contrast, a prominent feature of dreaming consciousness is that we neither engage in reflection, nor are we in control: we do not pause to assess our experiences or ask questions—we just go on dreaming, passively accepting whatever thought or image might come to mind without questioning it or intervening. We are unable

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to introspect, to realize that we are dreaming, to control our thoughts, and to imagine ourselves in the past or in the future[175]. This has been called the "single-mindedness" of dreaming[176]. We are also eminently "gullible," impaired in our ability to analyze situations intelligently, to question assumptions critically, to reason properly, and to make appropriate decisions. Holding contradictory beliefs is also quite common, and we easily accept impossible events or situations, such as flying. There is often uncertainty about orientation in space (where one is in the dream), about time (when the dream is taking place in personal history), and person (confusion about the gender, age, and identity of dream characters), yet we seem to accept them all. Indeed, dreaming has been described as confabulatory[177] because dreamers, like patients with spontaneous confabulations due to posterior orbitofrontal lesions[178], effortlessly invent narratives that do not correspond to reality while being partially based on real events. Dreaming has also been described as delusional, in the sense that fantastic events and characters are taken for real[179], pointing to a loss of "reality testing" and metacognition. A case can be made that all these "negative" symptoms stem from a suspension of reflection and perhaps more generally of cognitive access and control with their attendant processes, primarily working memory and attention.

The loss of metacognition and cognitive control during REM sleep dreaming has been attributed to a reduction of metabolic activity, compared to quiet wakefulness, in cortical regions well-known to support these functions[180,181], namely dorsolateral prefrontal cortex, frontopolar cortex, and inferior parietal lobule[182,183] (Fig. 5, but see[184]). The mechanisms underlying the reduced activation of much of prefrontal cortex during REM sleep are not known, though they may involve local changes in neuromodulation[185]. Alternatively, the intense intrinsic activation of limbic and posterior cortex during REM sleep may indirectly disrupt the engagement of recurrent interactions with prefrontal areas. The loss of metacognition and control during NREM sleep is easier to explain, given that prefrontal cortex shows abundant slow waves even when posterior cortex does not and subjects report dreaming[7].

Lucid dreaming—Remarkably, what is missing in typical dreams is also what makes "lucid dreaming" so special[186]. What distinguishes lucid dreaming is precisely the return of metacognition, with the ongoing realization that one is dreaming, rather than interacting with the world. Cognitive control, including the ability to remember at will and to steer the dream narrative in a chosen direction, such as flying, can also be restored. Lucid dreaming has some similarity to other unusual states, such as out of body experiences and sleep paralysis[187] (for the similarities between lucid dreaming to psychedelic states, see[188]). Lucid dreamers are also able to partially connect with the environment[189], responding to prearranged stimuli with prearranged motor responses (typically movements of the eyes, which are not paralyzed[190]) and they can remember what they experienced. Lucid dreaming has been extensively studied in the laboratory and used to demonstrate, for example, that dreamt experiences flow at slightly lower speed than waking ones (by signaling the time required by walking or counting)[191]. Stable lucid dreams only occur during REM sleep, especially in the early morning, accompanied by intense phasic phenomena. EEG studies have confirmed that lucid dreaming is not an intermediate state between wakefulness and sleep, but rather a highly activated form of REM sleep[192]. In a within-state case study

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with fMRI, when the dreaming subject was lucid compared to non-lucid, activity was higher over dorsolateral prefrontal and frontopolar regions, precuneus and inferior parietal lobule, and inferior and middle temporal gyri[193,194]. These regions of increased activation, broadly involved in cognitive and attentional control and metacognition, largely overlap with those deactivated in regular REM sleep compared to resting wakefulness. Moreover, lucid dreamers as a group show increased functional connectivity between left frontopolar cortex and angular and middle temporal gyri[195].

Consciousness forgetful

Considering that we are conscious throughout much of sleep for several hours every night, it is remarkable how little we remember our dreams. While a few people can report a dream nearly every night (high recallers), many hardly ever remember anything (low recallers). In fact, subjects exposed to the serial awakening paradigms are often surprised to discover that something was going on in their mind most of the time. If we do not commit the dream to memory or write it down immediately upon awakening, the story is lost forever, no matter how memorable or emotional.

This may not be too surprising for disjointed images or thoughts that have little context, as is also true of mind wandering during wakefulness. But narrative dreams are a different story. The same narrative, were it to happen when we are awake, would certainly leave a trace. On the other hand, learning about stories that did not happen in the real world would seem to be maladaptive. In fact, despite many attempts, there is no evidence that sleep can be exploited, in any practical way, to learn new skills, facts, stories, or languages[196], while there is substantial evidence that it can help in consolidating what has been learned[197,198].

Various possibilities have been considered to account for dream amnesia. The lack of reflection that characterizes dreams may be key[176]: if we are not actively involved in a goal-direct task, if we do not stop and notice, we may not engage neural circuits needed to encode new memories. A more general mechanism may be a change in neural plasticity associated with the suppression of monoaminergic signaling [199], especially during REM sleep[179]. Changes in the conditions for synaptic potentiation and depression, or in their balance, could explain why memory acquisition is suspended despite the strong activation of medial temporal lobe circuits that are critical for it during wakefulness. In fact, a transient increase in noradrenaline due to autonomic arousal or other causes may underlie the few dreams that are remembered in the morning. In addition, such micro-arousals might be responsible for sleep misperception—the feeling that one was not asleep that may occur even during deep slow wave sleep[6]. Finally, a transient increase in noradrenaline may also underlie the activation of midcingulate areas that distinguishes recalled dreams from white dreams after serial awakenings[83], perhaps by improving our ability to retrieve fleeting memory traces.

In summary, dreaming teaches us that consciousness can be present even in the near-absence of external connectedness, of cognitive control and metacognition, and without the accompaniment of memory for our experiences. In fact, it is instructive to consider dreaming consciousness not as an impoverished form of consciousness, but as the baseline or default condition. For the sleeper, each dreaming experience is real because, from one’s

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intrinsic perspective, experience is all there is. Wakefulness adds connectedness to the environment: the contents of experience, triggered by our interactions with the world, are sufficiently matched to it that we can behave adaptively (though sometimes, during daydreaming, connectedness is much reduced). Wakefulness also adds cognitive control and metacognition, which allow us to question bizarre experiences as "unreal," as well as the ability to control thoughts and actions. Finally, wakefulness adds the dependable companionship of memory, which preserves traces of much of what happens to us (though that, too, may be reduced during daydreaming).

Dreaming and the comprehensive sampling and recombination of conscious contents and narratives

To many, the most remarkable feature of dreaming consciousness is its rich, immersive quality and narrative organization, provided not by our interactions with the environment, but by endogenous sources. As aptly captured by the poet and playwright John Dryden, the loss of critical faculties in dreams is accompanied by the release of fantastic imagination: *Dreams are but interludes which fancy makes / When monarch reason sleeps, this mimic wakes. / Compounds a medley of disjointed things. A mob of cobblers and a court of Kings*[200].

Indeed, dreaming is like an immersive play improvised by an imaginative director, who picks a theme and recombines a set of well-rehearsed formulas to tell a story. But where is the director, and where does the story come from?

For simplicity, consider a typical REM dream as a narrative in which the successive episodes are tagged by a different spatial location. Such a story would be a mental navigation that takes the dreamer through a number of connected locations, each enriched with characters, objects, and actions. The scenes and actions experienced at each location may be a fantastic recombination of memories old and new, but their concatenation follows a narrative thread that usually has dramatic coherence and logic[177,201]. NREM dreams, being usually much shorter, are more likely to be exhausted by one or a few locations. In both cases, the substrate of consciousness would largely remain the same. The question is, what drives the scenes that are played out on that substrate, as well as their concatenation into a story?

Lesion and recording studies in humans

Lesion studies in humans who stop having story-like dreams but have preserved consciousness can shed light on the brain systems that are necessary for generating dream narratives, even though they are not necessary for being conscious. Most of these studies are based on retrospective self-reports, which makes it difficult to distinguish between loss of dreams and loss of memory for dreams. Nevertheless, they suggest that limbic regions may play a critical role. Individuals with bilateral lesions in ventromedial prefrontal cortex often report that they stopped dreaming[202], as do subjects who underwent leukotomies severing white matter tracts around ventromedial prefrontal cortex and affecting dopaminergic projections of the meso-limbic system[202,203]. Recently, a serial awakening paradigm was applied to four subjects with bilateral hippocampal lesions. The results show that these

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subjects had much reduced dreaming frequency and content, alongside with deficits in memory and imagination[204]. This is in contrast with individuals with bilateral basolateral amygdala lesions, who have no problem recalling dreams, although their dreams are shorter and simpler (they also have more positive and less negative emotions)[205]. Most subjects with bilateral striato-pallidal lesions who showed an auto-activation deficit during wakefulness—apathy, lack of spontaneous behavior and spontaneous thoughts (a "blank" mind)—reported no dreams when awakened in the laboratory; a third or so reported dreams that were rare, short, and simple[206]. Instead, lesions of dorsolateral prefrontal cortex do not abolish narrative dreaming[202].

One should also distinguish between loss of dreaming and loss of visual dreaming, which are often conflated. The latter, which is typically associated with loss of visual imagery during wakefulness, does not necessarily mean that all dreaming is lost Subject MX, a proficient imager, lost the ability to visualize following a cardiac procedure[207]. Perception was intact and, as expected, seeing faces activated visual regions, including the fusiform gyrus. While he continued to dream, his dreams lost the visual component and became more thought-like. Similarly, people who are blind from birth do dream, albeit not visually[31]. (For a discussion of dreaming, imagery, and the actions associated with dream imagery, see Box 3: Dreams and visual imagery).

Several neuroimaging studies have contrasted the difference in brain activation between REM sleep and resting wakefulness[157,208–212]. These contrasts, unlike within-state contrasts in NREM and REM sleep discussed above, are less suited to isolating the neural correlates of consciousness as such. However, considering that REM sleep is almost invariably associated with dreaming, they are informative about differences between the way the conscious brain functions in the awake and dreaming modes. The main findings of a meta-analysis based on six carefully selected studies[28] are presented in Fig. 5 (see also a previous meta-analysis[160]). As shown in the figure, there are several clusters of greater activation in REM sleep. These include high-level visual areas, such as the fusiform and lingual gyrus, in line with the vivid visual imagery of REM dreams. Also activated are limbic areas, including hippocampus, entorhinal cortex, and parahippocampal cortex, in line with the importance of memory sources for creating dream scenarios and stories. Whether retrosplenial cortex, which is critical for mental navigation, is also activated, is hard to say given the resolution of PET data. A portion of ventromedial prefrontal cortex, which may contribute to emotional and social situations common in REM dreams, is also activated, as is the anterior cingulate cortex. Finally, metabolic activity is higher in parts of brainstem, thalamus, basal forebrain, and basal ganglia (not shown).

Note also several clusters of deactivation in REM sleep, including the dorsolateral prefrontal cortex (see however[184]), rostral prefrontal cortex, orbitofrontal cortex, inferior frontal gyrus, middle and posterior cingulate cortex, precuneus, and inferior parietal lobule. As mentioned above, these deactivations are consistent with differences in cognitive functioning between REM sleep dreaming and waking consciousness[182].

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The REM sleep system in animal studies

While animal studies cannot provide direct evidence about dreaming, they have been invaluable in dissecting the brain circuits responsible for the triggering, maintenance, and manifestations of REM sleep.

What might be called the "REM sleep system" is a multi-level network of cell groups strongly activated during REM sleep that are found throughout the brain. Critical nodes are in the sublaterodorsal nucleus in the pons, which sends descending projections to the medulla. There, additional cell groups are responsible for the inhibition of motoneurons that causes the atonia of REM sleep, which prevents behavioral outputs despite intense brain activation. Other brainstem and lateral hypothalamic cell groups active during REM sleep are thought to inhibit noradrenergic, histaminergic, and orexinergic neurons, which is likely critical for sensory disconnection (the "brainstem gate"). Yet other brainstem populations promote the thalamocortical activation that underlies the low-voltage fast EEG activity of both REM sleep and wakefulness, with a primary role played by cholinergic systems.

A key node for the onset and maintenance of REM sleep is a group of melanin-concentrating hormone (MCH) neurons in the lateral posterior hypothalamus. This cell group is part of a brainstem-hypothalamic axis responsible for generating and entraining the theta rhythms that characterize REM sleep as well as exploratory behaviors during wakefulness[213,214]. Another key node is the supramammillary nucleus of the hypothalamus which, together with septal nuclei and the mesocortical dopaminergic system, leads to the activation of limbic regions[215].

Just like humans, rodents show a strong activation of many "limbic" regions during REM sleep, including dentate gyrus, medial entorhinal cortex, retrosplenial cortex, anterior cingulate, and medial prefrontal cortex, as indicated by the molecular marker Fos (reviewed in[172]). Rodent studies offer the further opportunity to dissect individual components of the REM system. For example, the dorsal dentate gyrus is activated by a set of supramammillary neurons involved in spatial behaviors, while another set, which projects to CA2 neurons, is involved in social behaviors[216]. The dentate gyrus, which contains many Fos positive neurons during REM sleep[217], may be central to the generation of concatenated activations. It can trigger the downstream activation of CA3 and of the superficial layer of CA1, which in turn projects to the deep layers of entorhinal cortex. By contrast, the deep layers of CA1 can be activated directly and via CA2 by the superficial layers of entorhinal cortex, and then project to prefrontal cortex[218,219]. These two CA1 pathways may have a different role in memory consolidation during sleep and memory acquisition during wakefulness. Specific populations in ventromedial prefrontal cortex (infralimbic cortex in rodents) are also activated and can facilitate the transition into REM sleep and promote its maintenance[220]. Other cellular populations highly active in REM sleep are found in medial entorhinal cortex and in the superficial layers of retrosplenial cortex (different from those activated by wakefulness)[64,221,222]. There, changing subsets of cells promote phasic events during REM sleep[63,64]. Retrosplenial cortex is critical for spatial navigation[223] and its coactivation with hippocampal circuits and anterior thalamic nuclei during REM sleep is important for memory consolidation[224]. (A small minority of claustrum neurons projecting

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to the superficial layers of retrosplenial cortex are also active in REM sleep, while the great majority are active in NREM sleep[225]).

The limbic activation profile that characterizes REM sleep must ultimately be due to changes in neuromodulators, primarily the high levels of acetylcholine (cholinergic innervation is stronger in limbic and paralimbic areas than in dorsolateral prefrontal cortex), the silence of monoaminergic systems, and perhaps burst firing within the meso-cortico-limbic dopaminergic system[226,227] that is typical of seeking behaviors[228].

Bottom-up, top-down, or inside-out, multi-level activation

It is generally accepted that, during waking perception, sensory stimuli drive activation along the sensory hierarchy in a bottom-up manner, whether the driving represents the stimulus signal (e.g.[229,230]) or the prediction error (e.g.[231,232]). After all, during wakefulness our experiences are very much triggered by external objects and events. The flow of features, objects, characters, scenes, events, and sequences of events that we experience is provided by the outside world through a succession of signals that percolate from the sensory periphery to primary cortical areas and from there to the rest of the brain, while engaging recurrent interactions with higher-level areas. In this sense, our experiences are triggered bottom-up. Bottom-up percolation would occur through cortical layer 4, employing both cortico-cortical and cortico-thalamo-cortical “driving” circuits[233].

How does activation propagate during dreaming? An early, bottom-up proposal was that, during REM sleep, cell groups in the brainstem would trigger PGO waves, which would randomly activate primary areas and percolate bottom-up across the sensory hierarchies[179,234,235].

Alternatively, higher-level areas might be the initial, top-down driver, activating lower-level ones through feedback connections that target superficial and deep layers, or perhaps just superficial layers, providing an “apical drive” to pyramidal neurons[144]. As already mentioned, many dreams have themes consistent with daytime preoccupations, such as social situations, personal goals, and fears of various kinds. Freud, for instance, claimed that dreams originate from unconscious wishes that are dressed up as images and concatenated into narratives[1]. In epileptic subjects, the direct electrical stimulation of associative cortices in the medial temporal lobe has occasionally induced dream-like events[236,237] (although subjects are simultaneously aware of their surroundings).

On the other hand, cortical activation may be triggered in parallel at multiple levels, in line with observations suggesting that cortical hierarchies may be much shallower than usually thought[238]. As we have seen, the REM sleep system is not a single generator, but an interconnected set of neuronal populations that span from the pons to limbic areas[150,172,173]. This is not surprising in a phylogenetic perspective, where new cell groups are progressively added on and reciprocally connected to pre-existing networks. If anything, the REM system would seem to be organized medio-laterally, or “inside-out,” since it involves tegmental and limbic areas on the medial surface of the brain, among them retrosplenial and ventromedial cortex. Both cortical regions seem able to trigger volleys of activation that spread to other cortical targets. This “inside-out” activation may in turn

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be facilitated by ascending inputs from the brainstem. For example, head direction signals from the brainstem can reach anterior thalamic nuclei[239] (presumably through the ventral tegmental-lateral mammillary pathway), and from there reach the retrosplenial cortex and the hippocampus[240]. The sawtooth waves of human REM sleep may also reflect this phasic, inside-out activation[61,62,241].

Even so, areas at higher levels of the cortical hierarchy would likely "anchor" the theme of a particular dream while lower levels firing patterns fluctuate more rapidly. Higher-level areas have longer "temporal receptive fields"[242,243] and divergent top-down activation can better select compatible firing patterns at lower levels than random activity converging bottom-up[244]. Moreover, during REM sleep trains of intrinsically generated activity patterns can be concatenated into long trains, since bursts of phasic events typically occur every 30-40 seconds in mice and 60-70 seconds in humans[60], possibly accounting for the single-mindedness of dreams. Wakefulness, instead, is likely to be interrupted by sensory inputs, behavioral requirements, or "reset" signals[147].

A theater metaphor may capture well the paradigmatic narrative dreams of REM sleep: the theme is usually set by high-level goals or fears, presumably through the contribution of ventromedial prefrontal cortex; the concatenation of episodes is a set of variations on the theme, provided by recombining "memories" indexed by the hippocampal formation —especially the dentate gyrus—and its interactions with retrosplenial cortex; and the immersive performance is played out, experience after experience, over the substrate of consciousness in posterior-central regions of the cortex.

Dream images and dream actions

It was suggested early on, based on anecdotal evidence, that the eye movements that characterize REM sleep might be related to the content of the dream (the so-called "scanning hypothesis"[245]). Recently, recordings of head direction cells in the anterodorsal thalamic nuclei of mice have shown that the coordination between head and eye movements is similar during the exploration of the environment in wakefulness and during REM sleep[239]. This finding reinforces the idea that action patterns during dreaming are usually congruent with the dreamt scene. Work with subjects affected by REM sleep behavior disorder has provided strong support for this notion[246]. These subjects, who lack the muscle atonia characteristic of REM sleep, usually present with strong muscle jerks—likely an exaggerated version of the muscle twitches that are visible in children. However, they may also behave as if fending off an offender, and occasionally act like smoking or singing[174]. When awakened, subjects report dream themes that are largely consistent with their behaviors (although a few subjects may deny having dreamt altogether[247]). Short episodes of standing, walking, feeding, attacking, or defending had previously been observed in cats with lesions to brainstem regions leading to REM sleep without atonia, and were attributed to the disinhibition of brainstem centers that control innate behaviors[248]. On balance, it is likely that the behaviors unmasked by the loss of atonia are controlled at multiple levels[249]. But what comes first, the images or the actions? Are the subjects with REM sleep behavior disorder "acting out dreams" or "dreaming out acts"[250]? During wakefulness, we may act in response to what we perceive, but we may also act and thereby change what we perceive

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(say, by performing exploratory eye movements). One possibility is that dreams may splice together memories that include both what we see and what we do, in line with the so-called ideomotor theory[251,252]. This possibility fits both with the passive nature of dreams—the lack of reflection and cognitive control—and with the inside-out, multi-level activation of REM sleep.

Dreams and the comprehensive recombination of intrinsic contents

A striking feature of dreams is the seemingly haphazard association of contents within individual experiences, and the frequently bizarre concatenation of scenes within a single narrative. The former has been characterized as "hyperassociativity," the latter as an expression of "divergent" thinking[179,253]. These prominent features have often prompted the intuition that dreams may serve to foster creativity, explore alternative strategies, and imagine future scenarios[254]. Indeed, a recent meta-analysis indicates that dreaming about a learning task is correlated with improved memory performance[255].

However, most dreams vanish from consciousness without leaving any trace upon which we may reflect It is thus unlikely that we dream to solve problems the way we would when awake, notwithstanding some well-known anecdotes, such as Kekule's dream of a snake biting its tail, which triggered his insight about the ring structure of benzene. Even previous events within a dream are soon forgotten and do not appear to influence the subsequent evolution of the dream. In fact, it has been argued that dreams may just be "spandrels"—byproducts of whatever the brain might be doing during sleep[183,256,257]. Or that REM sleep, and the associated dreams, may be a time for optimizing memory storage by strategically forgetting irrelevant aspects of what we have learned[258].

What is beyond doubt is that dreams offer a window on the brain's remarkable reshuffling of associations and concatenations engraved in its connectivity. Many studies have shown that dreams incorporate "residues" from waking experience—at least half of the time—often memories from about a week earlier (the "dream-lag" effect) and, especially towards the morning, memories from the distant past. However, even these recognizable elements are transformed, condensed, and recombined in new and unrelated contexts[259,260].

This recombination may seem problematic from an engineering point of view, but it makes eminent sense from a selectionist perspective. Selectionism, in the neural domain, assumes a pre-existing repertoire of connections and activity patterns that varies over time, rather than assuming a "blank slate" or a fixed set. It also assumes that encounters with the environment lead to the differential amplification of those variants that match the environment well[261,262]. Electrophysiological recordings, especially in the hippocampal system of rodents, have led to a better appreciation of the selectionist approach. This was prompted by the discovery of a large repertoire of ripple sequences that pre-exist encounters with new environments[263], undergo differential amplification during those encounters, and then go through a process of consolidation and reorganization that promotes a matching between intrinsic activity and connectivity and the environment[57,264,265].

The alternation of wakefulness and sleep adds an intriguing twist to this selectionist framework. Wakefulness, especially when associated with the exploration of a new

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environment, salient events, and goal-directed behavior, is the time for *up-selection*—the net strengthening of some pre-existing circuits based on ongoing activity. That ongoing activity is triggered and dictated by the specific features of the current environment and goals. By contrast, sleep is the time for *down-selection*—the net weakening of most synapses, except those that fit best with the pre-existing repertoire, in line with molecular, ultrastructural, and neurophysiological evidence[266]. Crucially, to be adaptive and promote the consolidation and integration of new memory traces, down-selection requires the comprehensive sampling and fluid recombination of the brain's intrinsic repertoire[267]. NREM sleep, when spontaneous activations are interrupted by OFF-periods, may be especially well-suited to the recombination of associations, representing a single episode or event, through the near-synchronous co-activation of a distributed activity pattern over the course of an ON-period. By contrast, REM sleep may be well-suited for the recombination of concatenations, representing a sequence of episodes or events, through the sequential occurrence of different activity patterns during prolonged tonic activations punctuated by phasic bursts.

This fluid recombination is precisely what sleep can provide that wakefulness cannot. During wakefulness, most of the time, we are necessary anchored to the here and now: we must act and react to the environment, whatever it offers us. Even when daydreaming, we mostly remain connected and monitor what is going on. Sleep, by enforcing the disconnection from the environment, the relinquishment of cognitive control and metacognition, and the suspension of memory acquisition, can instead enable the endogenous, comprehensive triggering and recombining of many different activity patterns. This is what makes sleep universal and irreplaceable. And dreams, seen this way, provide a phantasmagorical window on its fundamental function.

Conclusions

Perhaps the first and most important conclusion to be drawn from studies of consciousness in sleep and wakefulness is that most subjects, if asked to report what is going through their mind, report some kind of experience most of the time: almost without exception during wakefulness, and more than two thirds of the time during sleep.

Second, comparing brain activity when consciousness is present and absent within the same stage of sleep shows that loss of consciousness is associated with slow wave activity in a set of posterior-central cortical areas. Instead, prefrontal activation is important for behavioral arousal, metacognition, and cognitive control.

Third, loss of consciousness during sleep (and other conditions) is associated with neuronal bistability—the breakdown of causal links during slow wave OFF-periods—within the substrate of consciousness. On the other hand, as long as the substrate of consciousness is in a state of causal competence, any pattern of activity over it will support an experience.

Fourth, as long as causal links are enabled, it does not matter whether the activity pattern over the substrate of consciousness is triggered exogenously, as is usually the case during waking perception, or endogenously, as is mostly the case during sleep.

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Fifth, dream experiences are much like waking experiences even though the brain is largely disconnected on both the sensory and motor side. This suggests that what determines the quality of an experience must be accounted for intrinsically, by the substrate of consciousness in its current state, rather than extrinsically, by reference to sensory inputs or motor outputs. Of course, what the brain dreams of is ultimately due to a long history of interactions with the environment that have molded its causal connectivity through cycles of wakefulness and sleep.

Sixth, dreams show us what consciousness is like in the near absence of metacognition and cognitive control. Not only are dreaming experiences similar in content to those we have when we are in control, but they can be just as vivid and "realistic."

Undoubtedly, some of these conclusions are still controversial and many important issues are not yet resolved. What is responsible for the sensory and executive disconnection of the dreamer, especially during REM sleep? Which mechanisms are involved in the intrinsic triggering of dream experiences, especially for their concatenation into stories? And why do we not remember most of our dreams, including narrative ones?

Finally, not only can sleep serve as a window on consciousness, but consciousness can serve as a window on sleep. Dreaming consciousness indicates that the sleeping brain is far from quiescent. Instead, it samples states and sequences of states from the same repertoire from which the waking brain samples perceptions. In fact, freed from enslavement to the current environment, cognitive control, and metacognition, the sampling can be much more comprehensive. The systematic recombination of the brain's intrinsic causal repertoire revealed by our dreams, coupled with our lack of memory for them, suggests that sleep may be a time for the smart reorganization of that repertoire through down-selection.

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Box 1:

Sleep and arousability

The investigation of brain mechanisms of arousal begun with the discovery that electrical stimulation of the brainstem reticular formation could activate or "desynchronize" the synchronized EEG of anesthetized cats[118]. Since then, a number of "ascending reticular activating systems" have been identified, including noradrenergic, dopaminergic, cholinergic, serotoninergic, histaminergic, orexinergic and glutamatergic cell groups, which share two main features: diffuse projections to cortex and thalamus and higher firing during wakefulness than NREM sleep[268,269]. It was also discovered that, during REM sleep, cholinergic and dopaminergic systems resume firing, whereas the remaining systems become silent Among the latter, the noradrenergic system of the locus coeruleus is most closely linked to arousability from sleep: locus coeruleus stimulation invariably awakens mice from both NREM sleep and REM sleep within a few seconds[270], and optogenetic silencing of LC neurons reduces awakenings evoked by sounds[271]. Histaminergic neurons may promote the connection with the environment during cataplexy (reviewed in[146]). Among the cell groups active in REM sleep, optogenetic stimulation of dopaminergic neurons rapidly awakens mice from NREM sleep but much more slowly from REM sleep[227,272]. Stimulation of cholinergic neurons during NREM sleep either awakens mice or promotes the transition to REM sleep, preventing bistability, but has no effect during REM sleep[273,274].

When the brainstem reticular activating system was discovered, it was found that EEG activation could also be obtained by stimulating intra-thalamic nuclei (although, after thalamic lesions, the EEG could still be activated from the brainstem)[118]. We now know that arousal can be induced by stimulating some thalamic nuclei but not others. Thus, no arousal is obtained by stimulating the mouse ventral posteromedial nucleus that connects to primary somatosensory cortex[275], the mouse ventral medial geniculate nucleus that projects to primary auditory cortex[276], and the monkey dorsomedial nucleus that is mainly connected to prefrontal cortex[277]. On the other hand, stimulating nuclei rich in calbindin-positive "matrix" cells with broad cortical projections to layer 1, such as the mouse ventromedial nucleus, arouses mice from NREM sleep and anesthesia (but not from REM sleep)[275]. In monkeys, the electrical stimulation of the centrolateral nucleus, which projects to superficial and deep layers of frontal and parietal cortex, produces arousal from anesthesia[277,278]. In humans, stimulation of the central thalamus can produce behavioral improvement in severe disorders of consciousness[279]. These and other findings support the idea that a "mesocircuit" involving a fronto-striatal-pallidal-thalamic loop[280] may be critical for arousability.

Recently, it was shown that direct optogenetic activation of cortical neurons in posterior parietal cortex (and, less effectively, in medial prefrontal cortex) can produce cortical activation (decrease in low frequencies) and behavioral arousal (recovery of the righting reflex) from both NREM and REM sleep. Optogenetic cortical stimulation could also arouse mice from a coma-like state induced by the injection of GABA agonists in the midbrain and produce cortical EEG activation from deep sevoflurane-dexmedetomidine anesthesia[281].

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The initial demonstration of the reticular activating system was obtained in anesthetized animals. However, unlike anesthesia, sleep is a quiescent state of reduced responsiveness that is rapidly reversible through sensory stimuli[3], likely because peripheral inputs are sufficient to “awaken” brainstem activating systems. On the other hand, general anesthetics directly suppress neural excitability throughout the brain, including the reticular activating system and the mesocircuit, which would explain why arousability by sensory stimuli is lost. The lack of arousability during anesthesia may also be due to a deeper suppression of excitability in the mesocircuit As shown by a recent study using intracranial recordings and electrical stimulation, measures of neural complexity and connectivity are reduced during both deep NREM sleep and propofol anesthesia in posterior cortical regions, presumably reflecting loss of consciousness in both states. However, propofol leads to a much more profound reduction in prefrontal cortical areas, likely reflecting loss of arousability during anesthesia[282].

Rapid arousability distinguishes sleep not only from anesthesia, torpor, hibernation, and coma, but also from states of prolonged unresponsiveness with eyes open[283]. In many such cases, while subjects appear awake (eyes open), consciousness is most likely absent[140]. In a few instances, like patient 1 in[284], eyes-open unconsciousness was accompanied by continuous but purposeless movements of the head, trunk, and limbs. This was associated with relatively increased metabolic activity in anterior cortex and widespread destruction of parietal and occipital cortex. Conversely, as demonstrated by sleep, consciousness can be present in the absence of behavioral arousal. In fact, arousability is as low during phasic REM sleep, when subjects almost invariably dream, as it is during NREM sleep stage N3, which is often dreamless (reviewed in[146]). As discussed in the main text, low arousability is likely due to bistability in anterior cortex, whereas the absence of consciousness is associated with bistability in posterior-central cortex.

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Box 2:

Sleep onset experiences

EEG patterns vary rapidly during the transition from wakefulness to sleep[285,286](see also[287]). Systematic awakening during 9 codified substages of sleep onset[288] nearly always yield conscious reports[285,289,290]. Subjects are progressively more likely to report sensory and especially visual imagery up to substage 5. When interrupted, they frequently deny that they had been asleep (sleep misperception[6]). Some subjects may progress to show partial or full signs of REM sleep.

The advance from substage 1 to 5 is associated with a decrease in alpha activity (8-13 Hz) in the back of the brain and an increase in the front. Frontal regions also show a progressive increase in slower frequencies, suggesting that anterior regions may “go to sleep” before posterior regions[286]. This is consistent with neuroimaging studies showing increased metabolic activity in the visual cortex[291]. Consistent with the relative activation of posterior cortical areas, multivoxel decoding of fMRI activity patterns predicts the contents of visual imagery at sleep onset, especially from high-level visual areas[292].

A “microphenomenological” investigation of sleep onset can be a powerful tool for the study of consciousness[25]. It focuses on the dissection of experiential contents, their sources, and their sequencing, by exploiting “micro-dreams” often lasting just one second[170]. It also attempts to characterize the progressive shift from the “real world” to the “dream world,” and from fleeting to more elaborate experiences[25,170].

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Box 3:

Dreams and visual imagery

How does dreaming relate to waking imagery? Like dreaming, waking imagery involves many of the same brain areas that are activated by perception[293], although the strength of the activation is generally lower[294,295]. Decodability of visual contents during perception is highest in early visual cortex, comparable in perception and imagery in area V3, and highest during imagery in the intraparietal sulcus, which is involved in generating and maintaining stimulus-specific imagery signals[296]. Decoding during sleep onset dreaming is also best in higher-level visual areas[292].

During imagery, compared to perception, the flow of activation seems to reverse, going from high- to low-level areas[297,298]. Effective connectivity behaves similarly, and the vividness of imagery is correlated with top-down coupling[299]. There is also some evidence that imagery may rely primarily on top-down connections targeting infragranular layers, unlike perception or illusory completion[300], and may act by modulating neural activity, rather than driving it directly[295,301]. There is still no study comparing dreaming and imagination or examining the direction and cortical layer specificity of signal flow during dreaming.

Lesion studies suggest a similar substrate for visual imagery and visual dreaming (but not for dreaming as such, as mentioned in the main text). Initial case reports of loss of imagery and dreaming led to the designation of Charcot-Wilbrand syndrome[302]. Further case reports showed that individuals with posterior left hemisphere lesions reported loss of visual dreams and problems with imagery[303]. Conversely, most subjects with imagery problems after brain lesions also reported a loss or reduction of dreaming[304] (see also[202]). (However, a subject who stopped dreaming after a bilateral occipital artery infarction affecting the right inferior lingual gyrus and the right posterolateral thalamus, a loss confirmed with REM sleep awakenings, could imagine her surroundings[302]). More recent evidence concerning regions critical for image generation points specifically to part of the left fusiform gyrus, which may be well positioned to mediate interactions between language areas and high-level visual areas[305,306]. In line with such case reports, large surveys of subjects with self-reported aphantasia show a correlation between low or absent visual imagery and reduction of dreaming[307] (although the assessment of both dreaming and imagery from self-reports can be problematic). Visuo-spatial imagery is also the skill that is maximally correlated with dream recall in adults[308] and, in children, imagery and dream recall develop hand in hand[257].

On the other hand, there are major differences between dreaming and imagery, the most obvious being that dreaming, unlike imagery, is not only more vivid, but effortless, perhaps because it has little competition from sensory stimuli[309]. And of course, while we accept dreams as veridical, we are aware that waking imagery is not.

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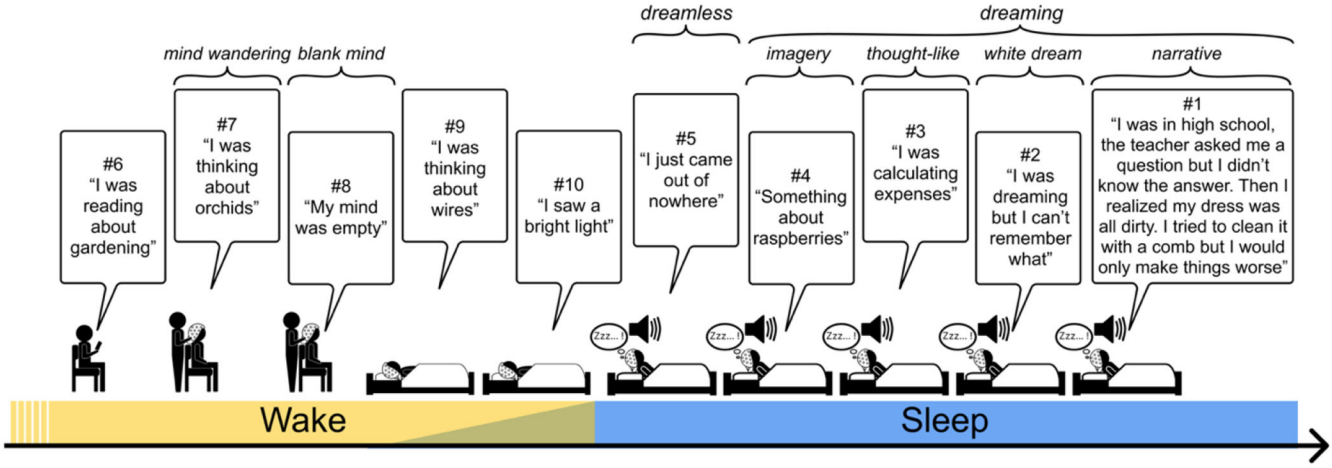

Figure 1: Experience sampling in wakefulness and sleep.

Examples of experience sampling at various times during the day, at sleep onset, and during sleep (serial awakening paradigm). The examples also introduce the nomenclature used throughout the review.

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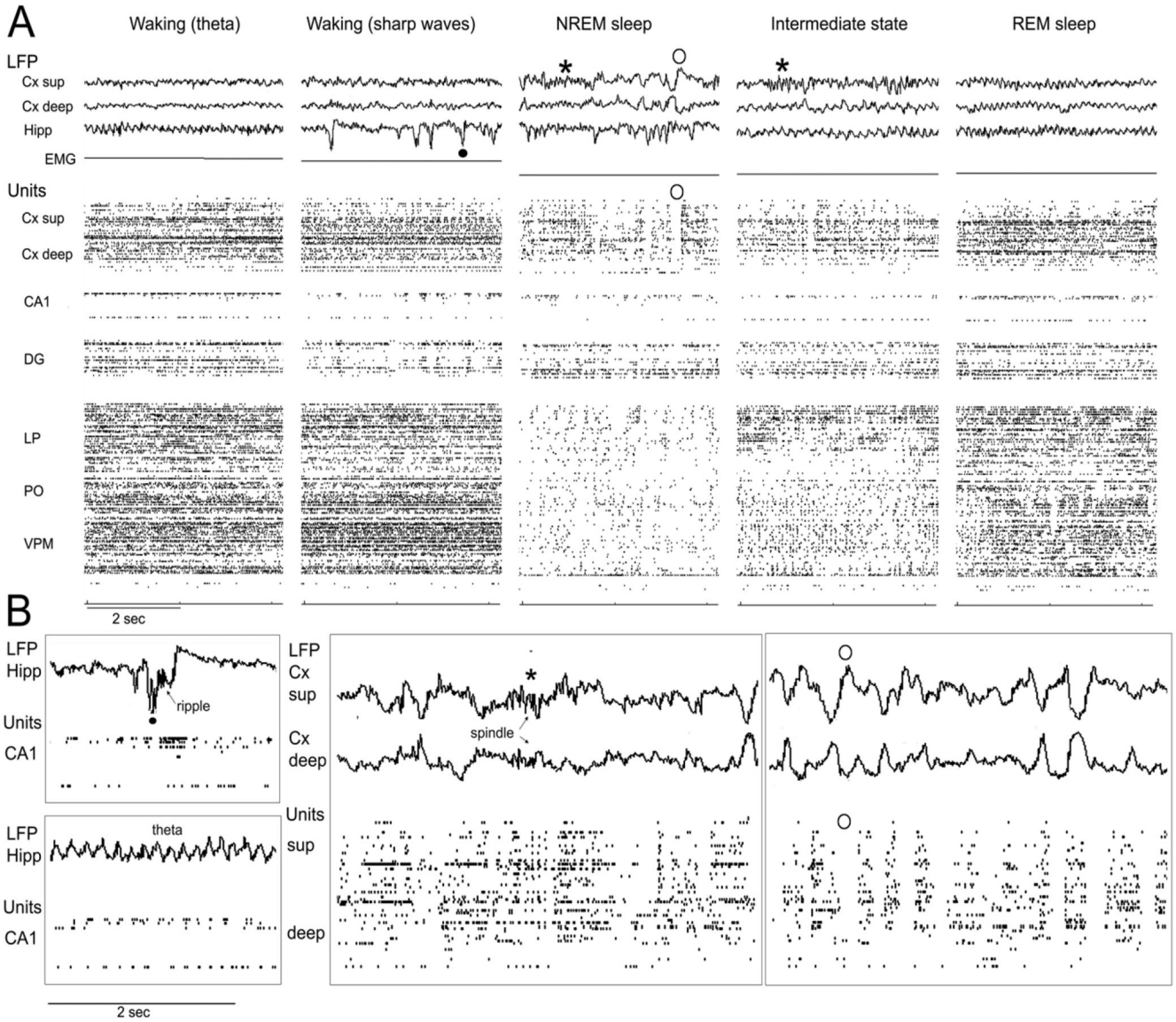

Figure 2. Neural activity in sleep and wakefulness.

A. Recording of local field potentials (LFP) and unit activity (units) in a freely moving rat during wakefulness and sleep, using a high density Neuropixels probe (384 sites) spanning, from dorsal to ventral, parietal association cortex (all layers), dorsal hippocampus (CA1 and dentate gyrus, DG), and multiple thalamic nuclei (lateral posterior/pulvinar, LP; posterior, Po; ventroposterior medial, VPM). Three LFP channels are shown at the top (superficial and deep cortex, CA1, stratum radiatum) for each behavioral state. EMG: electromyogram. Five behavioral states are shown, from left to right: active wakefulness with strong theta activity evident especially in the hippocampal LFP; quiet wakefulness with several hippocampal sharp waves (one indicated by a black filled circle); NREM sleep with a spindle (asterisk) and several slow waves (one indicated by the open circle) in the cortical LFPs and hippocampal sharp waves; intermediate state (transition from NREM sleep to REM sleep) with dense spindling in cortex and theta activity in the hippocampus; REM sleep with strong

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theta activity in both cortex and hippocampus. **B**. Major rhythms and oscillations are shown at higher magnification: sharp wave coupled to a ripple in the hippocampal LFP, associated with strong CA1 firing; hippocampal theta activity with rhythmic unit firing in CA1 and DG; spindle detected in superficial and deep cortical layer, and associated cortical firing; many cortical slow waves, one indicated by the open circle. Note the inverted polarity in superficial and deep layers, and the association with OFF periods (100-200 milliseconds of unit silence).

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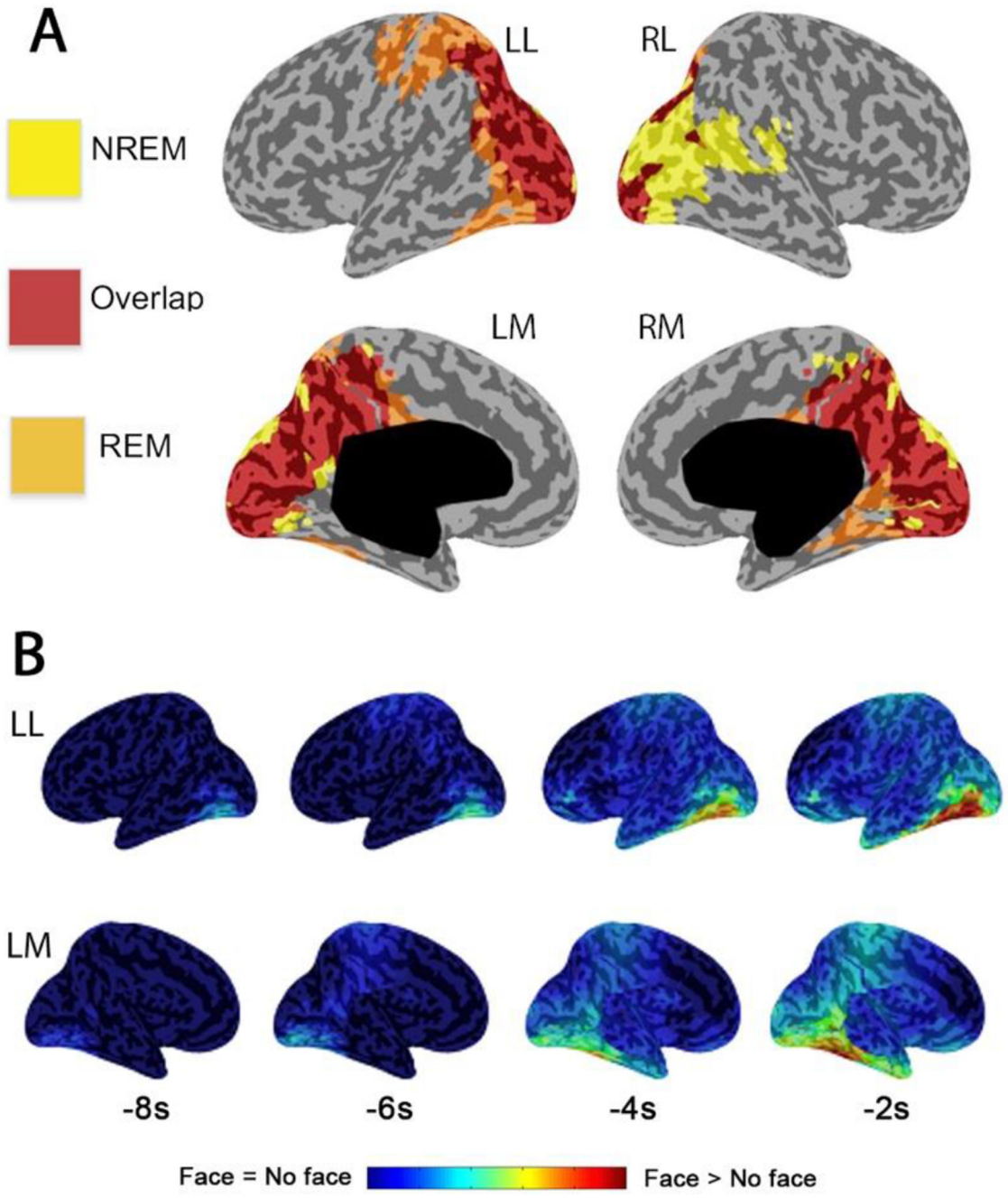

Figure 3: Neural correlates of loss of consciousness in sleep.

(A) Significant increase in slow wave activity (1-4 Hz) in subjects reporting no experiences vs. dreaming upon awakening within NREM sleep (in yellow) and within REM sleep (in orange). A conjunction analysis is shown in brown. The results are based on high-density EEG data reported in Siclari et al., 2017. The increase in slow wave activity during dreamless sleep is localized primarily over posterior-central cortical regions. LL (left lateral view), LM (left medial), RL (right lateral), RM (right medial). (B). Increased high-frequency activity (20-50 Hz) when subjects reported dreaming of faces, vs. dreaming of other

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contents, just before awakening from REM sleep. The increase in high-frequency activity is localized over the before awakening from REM sleep. The increase in high-frequency activity is localized over the fusiform gyrus and is maximal a few seconds before the awakening (from Siclari et al., 2017).

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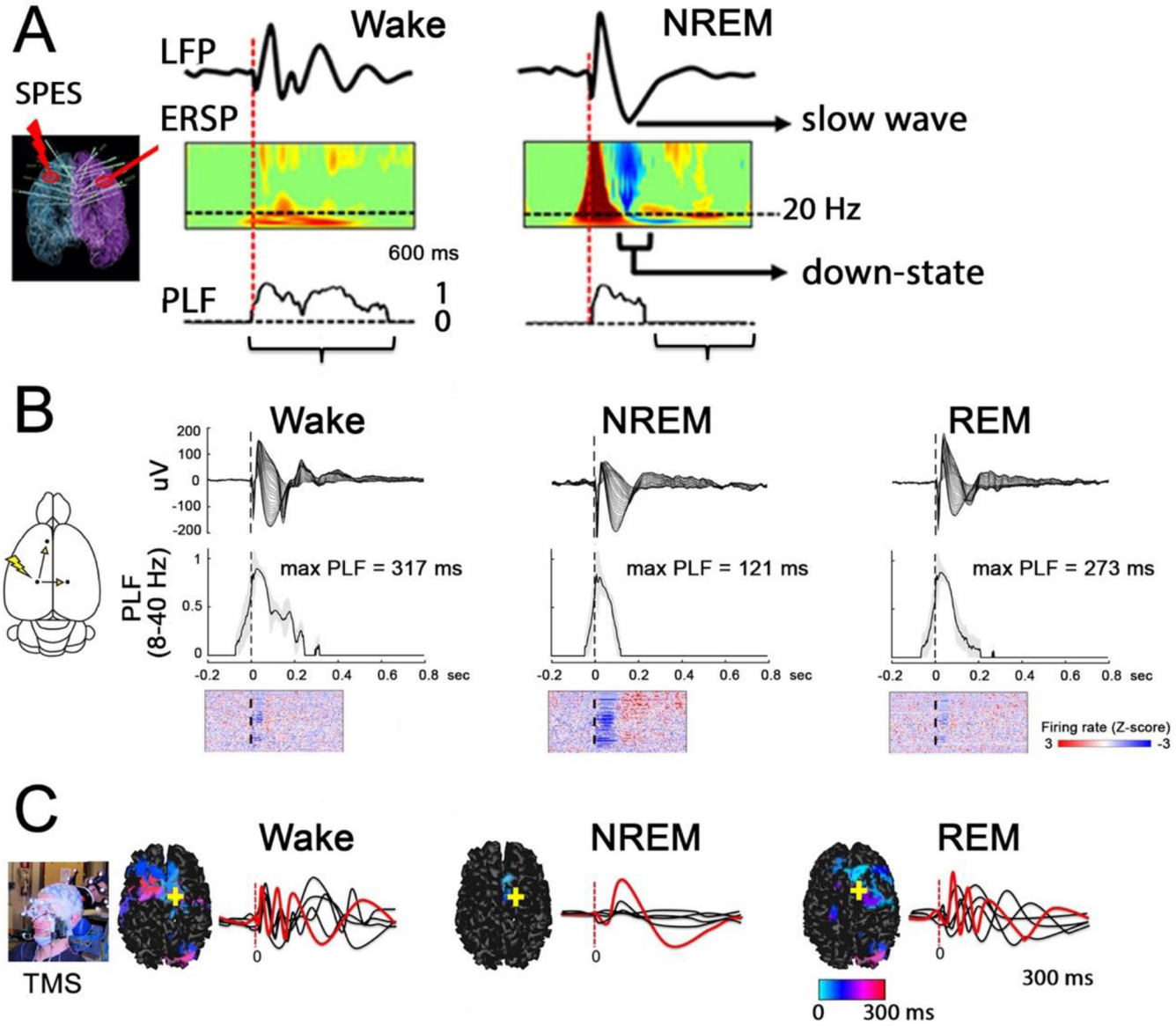

Figure 4: Neural mechanisms of loss of consciousness in dreamless sleep.

A. During wakefulness, single-pulse electrical stimulation (SPES) triggers a complex, prolonged response in the local field potential (LFP) recorded with simultaneous stereotactic EEG; This is also evident in the event-related spectral perturbation (ERSP); moreover, the phase-locking factor (PLF) is significantly elevated for more than 500 milliseconds, indicating sustained causal interactions triggered by the stimulus. During deep NREM sleep, SPES triggers an LFP response resembling a slow wave, associated with the suppression of high frequencies in the ESRP, and followed by the abrupt decay of the PLF (adapted from Pigorini et al., 2015). B. Schematic of the rat brain displaying the location of stimulating and recording probes in one rat (left) and examples of event related potentials (top), max phase locking factor in the 8-40 Hz range (middle) and single-unit peri-stimulus time histograms in a parietal probe locked to the onset of the electrical stimulation (bottom)

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across behavioral states in one rat. Units are sorted by depth. The vertical dashed line indicates the onset of the electrical stimulation (adapted from Cavelli et al., 2023). C. Spatiotemporal cortical current maps of TMS-induced activity during wakefulness, NREM, and REM sleep in a representative subject. The inset is the setup for TMS/EEG. For each significant time sample, maximum current sources are plotted and color-coded on the subject's MRI according to the activation latency (light blue, 0 milliseconds; red, 300 milliseconds). The yellow cross marks the TMS target on the cortical surface. To the right is the evoked response, in red for an EEG channel near the stimulation site, and in black for four representative channels further away (adapted from Massimini et al., 2010).

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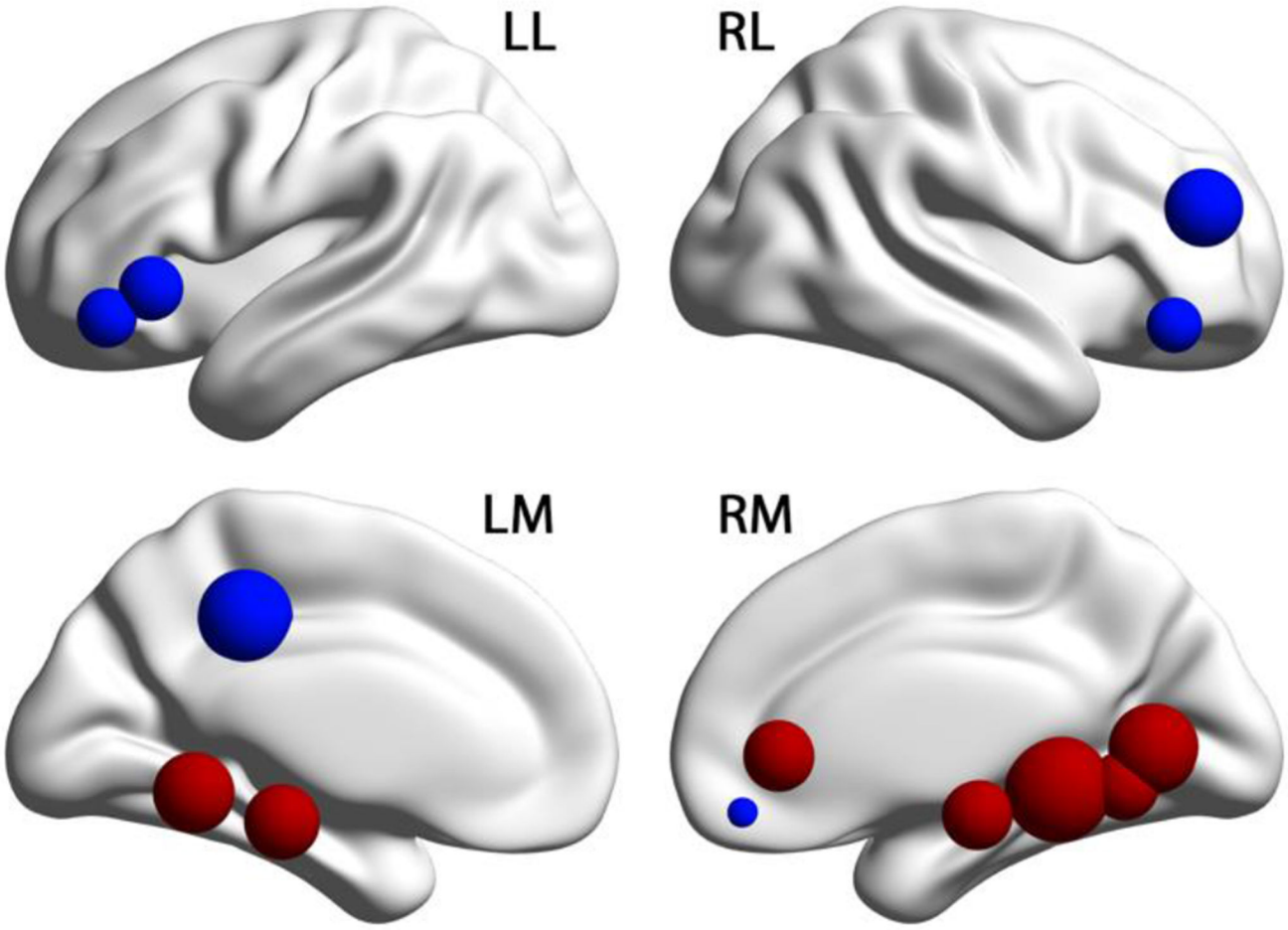

Figure 5: Comparison of brain activity between REM sleep and resting wakefulness.
Cortical areas more and less active during REM sleep compared to quiet wakefulness are plotted in red and dark blue, respectively. Significant clusters were obtained by (Fox et al., 2013) using an activation likelihood estimation analysis across 6 PET studies (Maquet et al., 1996; Nozfinger et al., 1997; Braun et al., 1997; Braun et al., 1998; Maquet et al., 2000; Peigneux et al., 2001) for a total of 61 subjects. The figure was created using a standard template (BrainNet Viewer (http://www.nitrc.org/projects/bnv/, Xia et al., 2013). Node locations centered on peak activation / deactivation coordinates as reported by (Fox et al., 2013), then projected to cortical surface with node sizes proportional to the number of voxels for each cluster. Subcortical structures that were more (pons, midbrain and caudate nucleus) or less activated (superior longitudinal fasciculus) during REM sleep than during wakefulness according to the (Fox et al., 2013) meta-analysis are not displayed in the figure.